

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Any 4

Total Marks:40

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work
and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

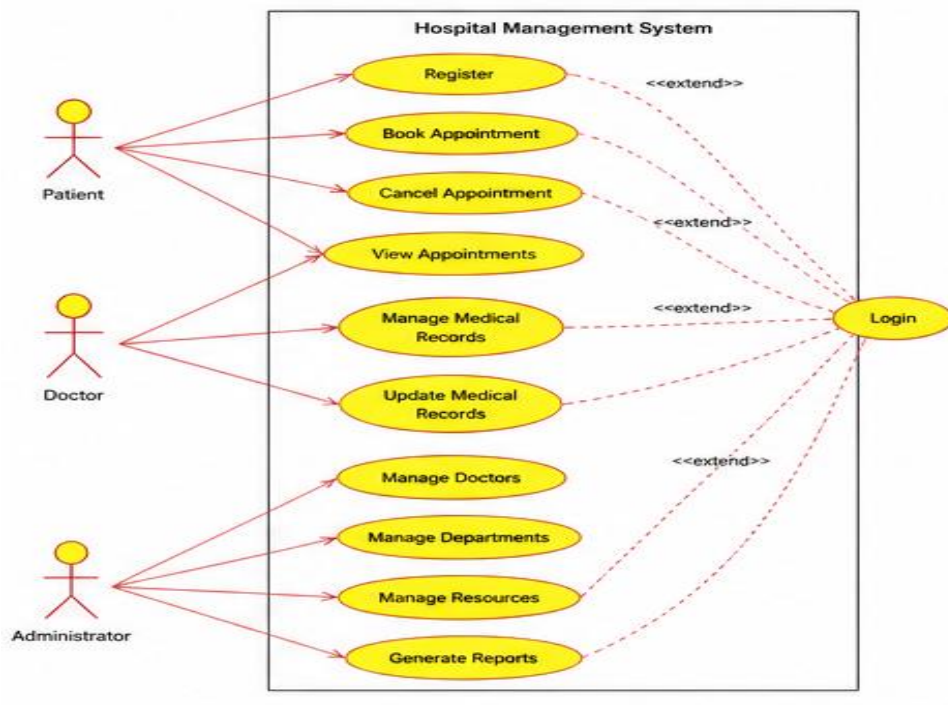
Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

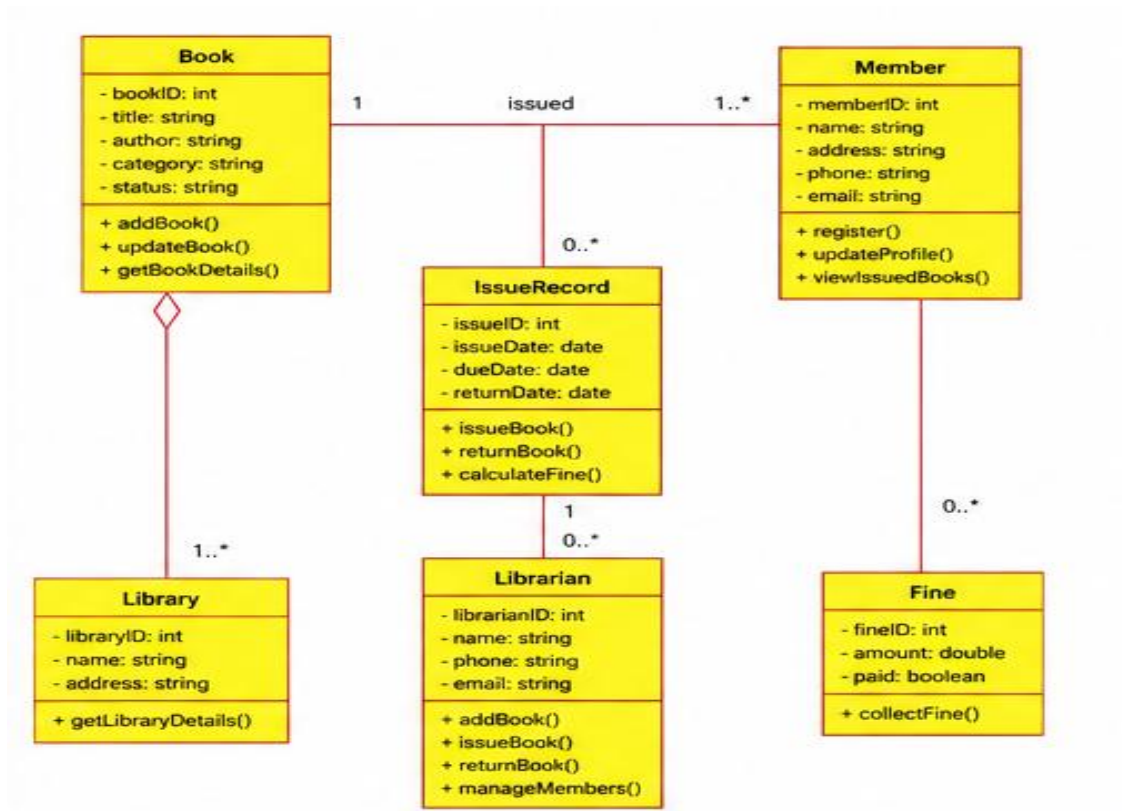
1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.



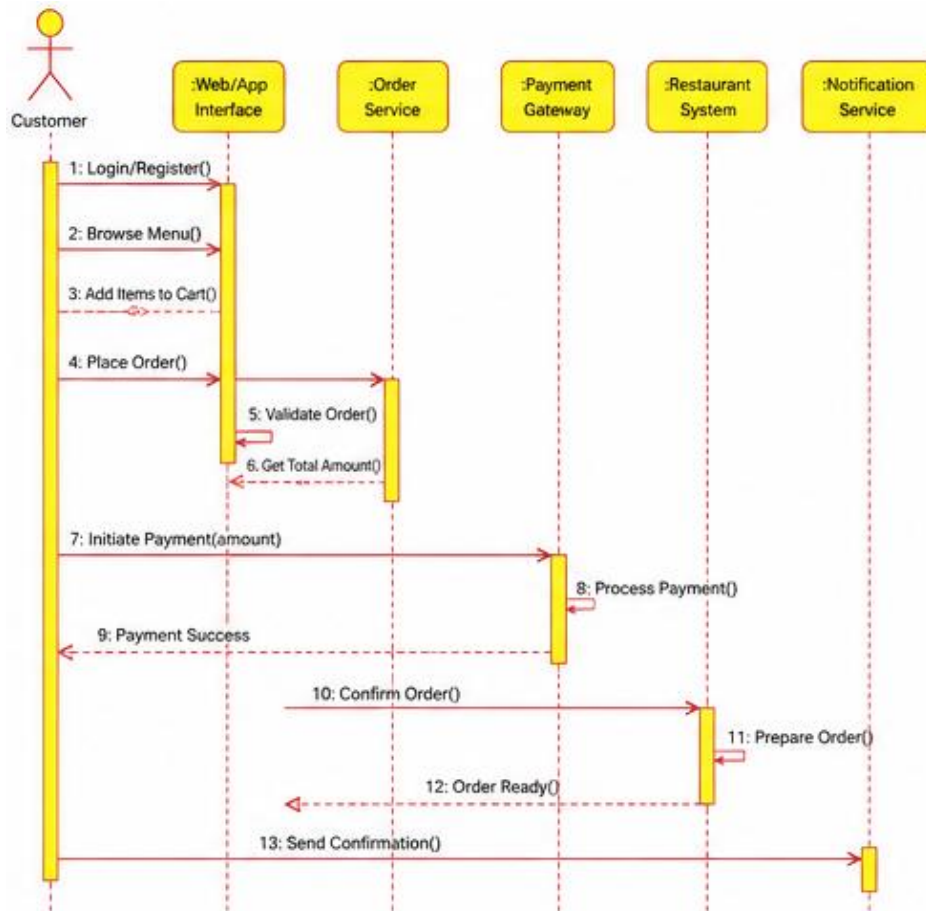
2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.



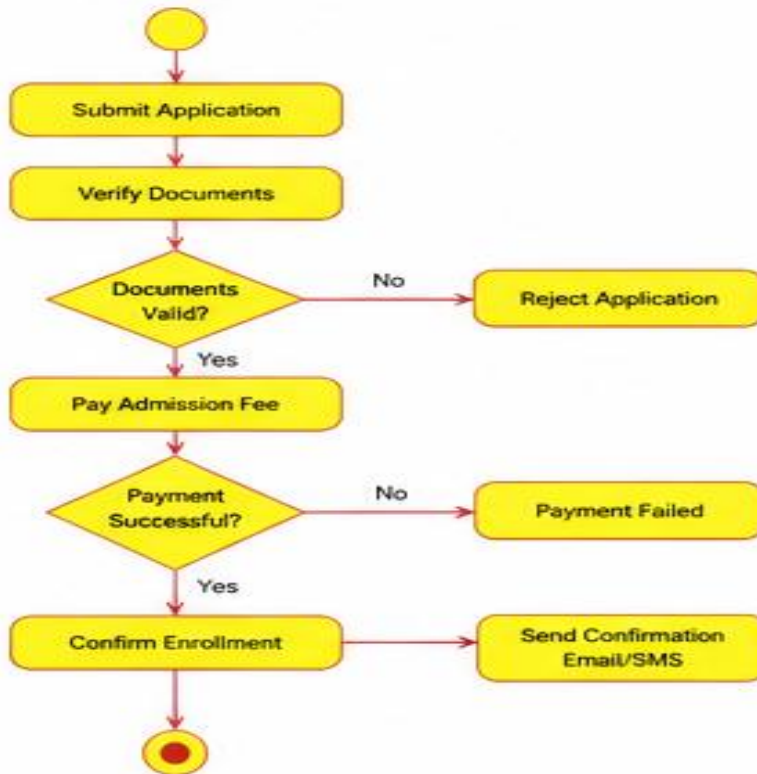
3. Sequence Diagram Problem

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.



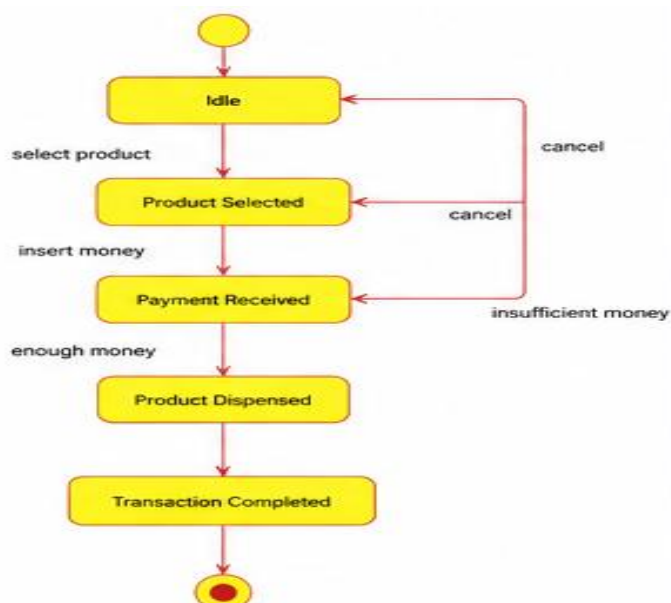
4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.



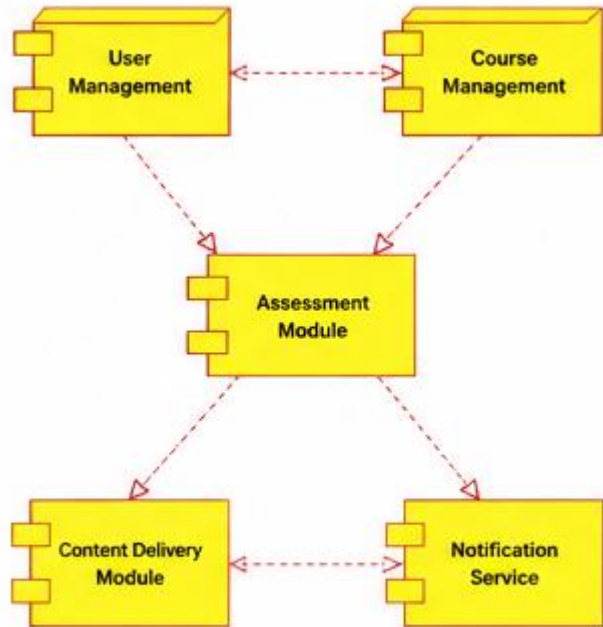
5. State Diagram Problem

A **Vending Machine System** operates through states such as Idle, Product Selected, Payment Received, Product Dispensed, and Transaction Completed.



6. Component Diagram Problem

An **Online Learning Platform** consists of modules such as User Management, Course Management, Assessment Module, Content Delivery Module, and Notification Service.



7. Deployment Diagram Problem

A **Smart Traffic Monitoring System** uses roadside sensors to collect vehicle data. The data is transmitted to a central cloud server where traffic authorities monitor congestion through a web dashboard. Alerts are generated during traffic violations and accidents.

